

Ibtasam Ur Rehman

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LinkedIn | GitHub | Google Scholar | ResearchGate

EDUCATION

Ho Chi Minh City University of Technology

Master of Science in Computer Science (CGPA: 7.54/10 $\approx$ 3.3/4.0)

Ho Chi Minh, Vietnam

Aug 2023 – June 2025

Riphah International University

Bachelor of Science in Software Engineering (CGPA: 2.94/4.0)

Islamabad, Pakistan

Aug. 2017 – Sept 2021

MASTER'S THESIS

Machine Learning Application Method to Detect Ophthalmic Diseases

Master's Thesis - Supervisor: Dr. Pham Hoang Anh

- Achieved 95.24% accuracy in cataract detection using an optimized RBF-SVM classifier, outperforming Random Forest (94.44%) and Polynomial SVM (92.86%) through extensive hyperparameter tuning with GridSearchCV.
- Engineered comprehensive feature extraction pipeline including HSV color features, Canny edge detection, and Hu moments for shape analysis, creating robust feature vectors for classification.
- Developed and preprocessed a dataset of 658 retinal images, implementing advanced techniques like unsharp masking and color space transformations to enhance model performance.
- Conducted comparative analysis of multiple ML algorithms, establishing RBF-SVM as the optimal model for ophthalmic disease detection with consistent 95% F1-score performance.

EXPERIENCE

The Millennium Universal College TMUC

Lecturer

Sept 2025 – Present

Islamabad, Pakistan

- Instructed and mentored students in a range of technical subjects including Mobile Computing, Full Stack Development, and Web Technologies
- Developed and delivered comprehensive course materials and hands on projects to provide students with foundational knowledge and practical skills.

Advanced Intelligence Technologies (AITech Lab)

Researcher

Feb 2024 – June 2025

Ho Chi Minh, Vietnam

- Conceptualized and developed advanced AI and IoT solutions, applying machine learning techniques to enhance system capabilities across various projects
- Collaborated with faculty and research teams to define project goals and identify knowledge gaps, contributing to literature reviews and aligning project objectives with real-world problems
- Applied machine learning algorithms to real-world data analysis tasks, optimizing models for specific challenges in AI. Worked on classification, prediction, and disease detection tasks across various projects

Ministry of Information Technology and Telecommunication

Senior Mobile Application Developer

Nov 2022 – May 2023

Islamabad, Pakistan

- Developed application with an intuitive UI, ensuring user-friendly features for easy adoption.
- Implemented secure API integrations using Flutter to maintain data confidentiality and integrity.
- Led iterative development cycles, gathering and incorporating feedback from stakeholders and users to continuously improve the app.

Techorra Tech

Mobile Application Developer

Aug 2021 – Nov 2022

Islamabad, Pakistan

- Performed in-depth user research to guide and optimize design decisions, ensuring alignment with user needs and expectations
- Developed high-performance cross-platform mobile applications using Flutter, focused on delivering seamless and intuitive user experiences.

Peer-Reviewed Conference Publication

1. **Cortex Vision: Detection of Ophthalmic Disease Using Machine Learning Algorithm.** In: *Smart Objects and Technologies for Social Good (GOODTECHS 2024)*, Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering (LNICST), vol. 649, Springer, Cham, 2025, pp. 138–149. *Ibtasam Ur Rehman, Hoang-Anh Pham*. DOI: 10.1007/978-3-032-01497-9_13
2. **Evolution of Explainable AI in Healthcare: Toward Trustworthy and Accurate Diagnostics.** Proceedings of the 2026 International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA), IEEE, 2026. *Ibtasam Ur Rehman, Abdulraqueeb Alhammadi, Jibran K. Yousafzai*. DOI: 10.1109/ACDSA67686.2026.11467739
3. **Hybrid Machine and Deep Learning in Brain Tumor MRI Analysis.** Proceedings of the 2026 International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA), IEEE, 2026. *Ibtasam Ur Rehman, Abdulraqueeb Alhammadi, Jibran K. Yousafzai, Sobia Asghar*. DOI: 10.1109/ACDSA67686.2026.11467573
4. **Convolutional Neural Network Pipeline with Feature Synthesis for Cardiovascular Risk Prediction.** In: *2026 IEEE Conference on Artificial Intelligence (CAI)*, IEEE, 2026, pp. 2009–2014. *Ibtasam Ur Rehman, Jibran K. Yousafzai, Abdulraqueeb Alhammadi, Muhammad Ikram, Maheen Ishfaq*. DOI: 10.1109/CAI68641.2026.11536599
5. **Lightweight Machine Learning Models for Pneumonia Detection in Chest Radiographs.** In: *2026 IEEE Conference on Artificial Intelligence (CAI)*, IEEE, 2026, pp. 2015–2020. *Ibtasam Ur Rehman, Jibran K. Yousafzai, Abdulraqueeb Alhammadi, Muhammad Ikram, Zoha Farooq, Maheen Ishfaq*. DOI: 10.1109/CAI68641.2026.11536642

Accepted Conference Papers

1. **SmartGluco: A Mobile Health Solution for Diabetes Risk Assessment Using Machine Learning.** Accepted and Presented at 4th International Conference on Innovations in Data Analytics (ICIDA 2025), Springer LNNS Series. *Ibtasam Ur Rehman, Muhammad Islam, Neha Janu, Preeti Narooka, Ahsan Fiaz, Zakaria Alomari*. [View publication](#)
2. **Deep Learning Architectures for Urolithiasis Classification: A Comparative Analysis of DNN, MLP and Autoencoder-based Models.** Accepted and presented at Malaysian Automatic Control Symposium (MARS2025). *Ibtasam Ur Rehman, Abdulraqueeb Alhammadi, Jibran K. Yousafzai*. [View publication](#)
3. **Hybrid AI Approaches for Clinical Diagnostics: A Case Study on Meningitis.** Accepted and presented at Malaysian Automatic Control Symposium (MARS2025). *Ibtasam Ur Rehman, Abdulraqueeb Alhammadi, Jibran K. Yousafzai*. [View publication](#)
4. **GELU-Activated Neural Network for Cardiovascular Risk Assessment: A Feature Engineered Deep Learning Approach Using Clinical Biomarkers.** Accepted and Presented at 2nd International Conference on Intelligent Systems and Security (ICISS2026). *Ibtasam Ur Rehman, Muhammad Islam, Stuti Pandey, Preeti Narooka*. [View publication](#)

Journal Papers (Under Review / Submitted)

1. **An Advanced Stacking-based Machine Learning and Deep Learning Framework for Breast Cancer Prediction.** Under review at *Big Data Journal* (2026). *Ibtasam Ur Rehman, Muhammad Islam, Basharat Hussain*. [View preprint](#)
2. **Predictive Analysis of Asthma Patients Using Vision Transformer (ViT).** Submitted to *Evolving Systems (Springer)* (2026). *Sana Tameer, Muhammad Ehsan Rana, Ibtasam Ur Rehman*. [View manuscript](#)

3. **Integrating Retrieval Augmented Generation with LLMs for Robust and Explainable AI Text Detection.** Submitted to *Big Data Journal* (2026). *Ibtasam Ur Rehman, Muhammad Islam, Muhammad Yousaf Rehman, Basharat Hussain.* View manuscript

Manuscripts in Preparation

1. **From Black Box to Transparent Diagnosis: The Role of Explainable AI for Predictive Diagnostics in Healthcare.** Manuscript in preparation; targeting a Q1 journal (*Machine Learning and Knowledge Extraction*). *Ibtasam Ur Rehman, Jibran K. Yousafzai, Abdulraqueb Alhammadi.* View manuscript

TECHNICAL SKILLS

Programming Languages: Python, Dart
Frameworks: Flutter, Material UI, FlaskAPI
Tools: Git, Android Studio, Visual Studio, PyCharm, Figma, Adobe
Libraries: Pandas, NumPy, Matplotlib, Plotly, CV2, Scikit-learn
Design Skills: UI/UX, Application Design, Wireframing, Prototyping, Visual Design, Interaction Design

ACADEMIC REFERENCES

Dr. Pham Hoang Anh Vice Dean Faculty of Computer Science HCMC University of Technology anhpham@hcmut.edu.vn	Dr. Musharraf Ahmed Assistant Professor & HOD Faculty of Computing Riphah International University musharraf.ahmed@riphah.edu.pk	Assoc. Prof. Dr. Nam Thoai Head of HPC Lab Faculty of Computer Science HCMC University of Technology namthoai@hcmut.edu.vn
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RESEARCH COLLABORATOR

Abdulraqueb Alhammadi Senior Lecturer, Faculty of Artificial Intelligence Universiti Teknologi Malaysia abdulraqueb.alhammadi@utm.my	Dr. Jibran Khan Yousafzai Assistant Professor College of Eng. & Applied Sciences American University of Kuwait jyousafzai@auk.edu.kw	Dr. Basharat Hussain Assistant Professor Faculty of Computing FAST NUCES, Islamabad basharat.hussain@isb.nu.edu.pk
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